



SCIENCE, TECHNOLOGY & SOCIETY

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GENERAL CONCEPTS AND HISTORICAL DEVELOPMENT

- Concepts & Definitions
- Historical Antecedents
- Intellectual Revolution

CONCEPTS AND DEFINITIONS

“Any sufficiently advanced technology is indistinguishable from magic”

- Arthur C. Clarke

CONCEPTS AND DEFINITIONS

What is Science?

- a **systematic process** of acquiring knowledge about the natural world through **observation** and **experimentation**
- the **pursuit** and **application** of knowledge and understanding of the natural and social world following a systematic methodology based on evidence

CONCEPTS AND DEFINITIONS

What is Technology?

- the **practical application** of scientific knowledge to create tools, systems, and processes that help humans **achieve specific goals, solve problems,** and **enhance** their lives

CONCEPTS AND DEFINITIONS

What is Society?

- refers to a **complex web** of **human interactions** and **relationships** that exist within a **specific geographic** area or among a group of people who share **common** interests, values, and norms.

HISTORICAL ANTECEDENTS

Ancient Period: Early Civilizations (Sumerians, Egyptians, Greeks, etc.):

- Cuneiform and Hieroglyphs
- Watermill
- Irrigation
- Potter's Wheel
- Papyrus
- Optical Lenses
- Mathematics, Astronomy, Engineering and Philosophy

HISTORICAL ANTECEDENTS

Middle Ages: Influence of Islamic, Chinese, European Civilizations

Technological Advancements:

- Gunpowder
- Compass
- Paper
- Paper Money
- Windmill

HISTORICAL ANTECEDENTS

Middle Ages:

Scientific Advancements:

Arab scholars preserved and translated Greek and Roman texts, contributing to the advancement of mathematics, astronomy, medicine and engineering.

Social and Cultural Contexts:

The Middle Ages saw the rise of universities and the development of new technologies

HISTORICAL ANTECEDENTS

Modern Period: Renaissance Period

The "cultural rebirth" following the Middle Ages, which saw the rediscovery of classical Greek and Roman philosophy, literature, and art.

This period fostered innovations in the arts, sciences, and politics under the philosophy of humanism.

HISTORICAL ANTECEDENTS

Modern Period: Renaissance Period

Scientific and Technical Revolution:

- Figures like Copernicus, Galileo, da Vinci, Kepler, Newton, Vesalius and Gutenberg made groundbreaking discoveries in Astronomy, Physics, Anatomy, Physiology Chemistry, Engineering and Mathematics

Art, Philosophy and Humanity:

- Artist like Leonardo da Vinci, Michaelangelo, Raphael, Donatello, Copernicus, and Philosophers such as Descartes, Erasmus, Machiavelli, Tomas More, Francis Bacon and Alberti

HISTORICAL ANTECEDENTS

Modern Period: Renaissance Period

Technological Ideas and Advancements:

- Microscopes and Telescopes
- Printing Press
- Mechanical Clocks
- Matches
- Pencils
- Submarines and Airplanes